

Nandhini MS

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WEEK 2

Module 5 - Python Backend Frameworks – Django, Flask & Fast API

HANDS-ON 3

TASK 1

Task 1: Serializers and Basic API Views

Goal: Create DRF serializers and your first API views

Step 1: Install Django REST Framework

Open the VS Code terminal and run:

```
pip install djangorestframework
```

After installation, you should see something like:

```
Successfully installed djangorestframework-3.xx.x
```

Step 2: Add DRF to settings.py

Open

```
coursemanager/settings.py
```

Find

```
INSTALLED_APPS = [
```

Add

```
'rest_framework',
```

Example

```
INSTALLED_APPS = [  
    'django.contrib.admin',  
    'django.contrib.auth',  
    'django.contrib.contenttypes',  
    'django.contrib.sessions',  
    'django.contrib.messages',  
    'django.contrib.staticfiles',  
  
    'courses',  
    'rest_framework',  
]
```

Save (**Ctrl+S**)

MY WORKFLOW

```
Lib\site-packages (from django) (3.11.1)
Requirement already satisfied: sqlparse>=0.5.0 in C:\Users\HP\AppData\Local\Python\pythoncore-3.14-64\
\Lib\site-packages (from django) (0.5.5)
Requirement already satisfied: tzdata in C:\Users\HP\AppData\Local\Python\pythoncore-3.14-64\Lib\site
-packages (from django) (2026.2)
PS C:\Users\HP\OneDrive\Desktop\Django project> python -m pip install djangoestframework
Python was not found; run without arguments to install from the Microsoft Store, or disable this shor
tcut from Settings > Apps > Advanced app settings > App execution aliases.
PS C:\Users\HP\OneDrive\Desktop\Django project> cd C:\Users\HP\coursemanager
PS C:\Users\HP\coursemanager> python manage.py runserver
Python was not found; run without arguments to install from the Microsoft Store, or disable this shor
tcut from Settings > Apps > Advanced app settings > App execution aliases.
PS C:\Users\HP\coursemanager> |
```

```
EXPLORER
DJANGOPROJECT

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS
Microsoft Windows [Version 10.0.26100.2894]
(c) Microsoft Corporation. All rights reserved.

C:\Users\visha\OneDrive\Desktop\DjangoProject>pip install django
Collecting django
  Downloading Django-5.1.6-py3-none-any.whl.metadata (4.2 kB)
Collecting asgiref<4,>=3.8.1 (from django)
  Using cached asgiref-3.8.1-py3-none-any.whl.metadata (9.3 kB)
Collecting sqlparse>=0.3.1 (from django)
  Using cached sqlparse-0.5.3-py3-none-any.whl.metadata (3.9 kB)
Collecting tzdata (from django)
  Downloading tzdata-2025.1-py2.py3-none-any.whl.metadata (1.4 kB)
Downloading Django-5.1.6-py3-none-any.whl (8.3 MB)
8.3/8.3 MB 9.1 MB/s eta 0:00:00
Using cached asgiref-3.8.1-py3-none-any.whl (23 kB)
Using cached sqlparse-0.5.3-py3-none-any.whl (44 kB)
Downloading tzdata-2025.1-py2.py3-none-any.whl (346 kB)
Installing collected packages: tzdata, sqlparse, asgiref, django
Successfully installed asgiref-3.8.1 django-5.1.6 sqlparse-0.5.3 tzdata-2025.1

[notice] A new release of pip is available: 24.3.1 -> 25.0
[notice] To update, run: python.exe -m pip install --upgrade pip

C:\Users\visha\OneDrive\Desktop\DjangoProject>django-admin startproject |
```

Django project runned sucessfully

Task 1

Step 26

Create a new file

courses

serializers.py

Paste this:

```
from rest_framework import serializers
from .models import Department, Course, Student,
Enrollment

class DepartmentSerializer(serializers.ModelSerializer):
    class Meta:
        model = Department
        fields = '__all__'

class CourseSerializer(serializers.ModelSerializer):
    class Meta:
        model = Course
        fields = '__all__'

class StudentSerializer(serializers.ModelSerializer):
    class Meta:
        model = Student
        fields = '__all__'
```

```
class EnrollmentSerializer(serializers.ModelSerializer):
    class Meta:
        model = Enrollment
        fields = '__all__'
```

Save.

Step 27

Open

`courses/views.py`

Replace everything with:

```
from rest_framework.views import APIView
from rest_framework.response import Response
from rest_framework import status
```

```
from .models import Course
from .serializers import CourseSerializer
```

```
class CourseListView(APIView):

    def get(self, request):
        courses = Course.objects.all()
        serializer = CourseSerializer(courses, many=True)
        return Response(serializer.data)
```

```
def post(self, request):
    serializer = CourseSerializer(data=request.data)

    if serializer.is_valid():
        serializer.save()
        return Response(serializer.data,
status=status.HTTP_201_CREATED)

    return Response(serializer.errors,
                    status=status.HTTP_400_BAD_REQUEST)
```

Save.

Step 28

Add this **below** the previous class.

```
class CourseDetailView(APIView):

    def get_object(self, pk):
        try:
            return Course.objects.get(pk=pk)
        except Course.DoesNotExist:
            return None

    def get(self, request, pk):
        course = self.get_object(pk)

        if not course:
```

```
        return
    Response(status=status.HTTP_404_NOT_FOUND)

    serializer = CourseSerializer(course)
    return Response(serializer.data)

def put(self, request, pk):
    course = self.get_object(pk)

    if not course:
        return
    Response(status=status.HTTP_404_NOT_FOUND)

    serializer = CourseSerializer(course,
    data=request.data)

    if serializer.is_valid():
        serializer.save()
        return Response(serializer.data)

    return Response(serializer.errors,
        status=status.HTTP_400_BAD_REQUEST)

def delete(self, request, pk):
    course = self.get_object(pk)

    if not course:
        return
    Response(status=status.HTTP_404_NOT_FOUND)

    course.delete()
```

```
        return  
        Response(status=status.HTTP_204_NO_CONTENT)  
  
    Save.
```

Step 29

Open

`courses/urls.py`

Replace with

```
from django.urls import path  
from .views import CourseListView, CourseDetailView  
  
urlpatterns = [  
    path('courses/', CourseListView.as_view()),  
    path('courses/<int:pk>/', CourseDetailView.as_view()),  
]
```

Now open

`coursemanager/urls.py`

Make sure it looks like this:

```
from django.contrib import admin  
from django.urls import path, include
```



```
urlpatterns = [  
    path('admin/', admin.site.urls),  
    path('api/', include('courses.urls')),  
]
```

Step 30

Run the server:

```
python manage.py runserver
```